

# Exploring the Toxicological Impact of Bisphenol A on Acne Vulgaris via the PLA2G2A–Arachidonic Acid Pathway

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**Objective:** To investigate the potential molecular mechanisms underlying the contribution of environmental bisphenol A (BPA) exposure to acne pathogenesis, with a specific focus on the PLA2G2A–arachidonic acid pathway.

**Methods:** We integrated network toxicology and machine learning to identify common target genes linking BPA exposure and acne. Core genes and key pathways were screened through differential expression, gene enrichment, and feature selection algorithms. The findings were validated in independent datasets and further investigated using immune infiltration analysis, molecular docking, and molecular dynamics simulations.

**Results:** Three core genes were identified, with PLA2G2A being the central hub. Functional enrichment analysis revealed significant involvement of these targets in arachidonic acid metabolism, PI3K-AKT signaling, and IL-6/JAK-STAT3 pathways. PLA2G2A was upregulated in acne lesions and positively correlated with pro-inflammatory immune cell infiltration. Molecular docking and dynamics simulations indicated stable binding between BPA and PLA2G2A.

**Conclusion:** These findings suggest that BPA exposure may contribute to acne development by targeting PLA2G2A, disrupting arachidonic acid metabolism, and modulating the local inflammatory microenvironment. Crucially, this study highlights the environmental risk of endocrine-disrupting chemicals in dermatology and identifies PLA2G2A as a novel, potential therapeutic target for comprehensive acne management.

**Keywords:** bisphenol A, acne vulgaris, PLA2G2A, arachidonic acid metabolism, network toxicology, endocrine disrupting chemicals

## Introduction

Acne vulgaris is a common chronic inflammatory skin disorder that primarily affects the pilosebaceous units of the face and trunk.<sup>1,2</sup> Its pathogenesis is fundamentally linked to the sebaceous gland, where excessive sebum production (seborrhea) driven by hormonal stimulation and altered lipid metabolism creates a microenvironment conducive to abnormal follicular keratinization and microbial colonization.<sup>1,2</sup> Epidemiological data indicate that the disease affects approximately 9% of the global population, with an incidence as high as 85% among adolescents aged 12 to 24 years, and still nearly 50% among individuals aged 20 to 29.<sup>3</sup> It exerts a significant impact on the physical and psychological health as well as the appearance of adolescents.

Endocrine-disrupting chemicals (EDCs) are a class of exogenous compounds capable of affecting the human hormonal system and disrupting endocrine functions, raising widespread concerns about their potential risks to public health.<sup>4,5</sup> These substances are commonly present in consumer products, plasticizers, pharmaceuticals, and aquatic environments.<sup>6</sup> Data from the U.S. National Health and Nutrition Examination Survey (NHANES) revealed a high

detection rate of various EDCs, including bisphenol A (BPA), in the general population. Notably, BPA was detected in 95.7% of urine samples collected during 2003–2004<sup>7</sup> indicating its extensive human exposure.

While acne is a multifactorial disease fundamentally driven by genetic predisposition, endogenous hormonal fluctuations, and microbial colonization,<sup>8</sup> the compounding impact of exogenous environmental factors is gaining increasing attention. Emerging epidemiological evidence suggests that exposure to environmental pollutants, including airborne particulate matter and EDCs, is significantly associated with the exacerbation and increased prevalence of acne vulgaris.<sup>9</sup> Existing evidence suggests that EDCs can interfere with the levels and signaling pathways of these key hormones through various mechanisms, potentially affecting the physiological activity of sebaceous glands and contributing to the pathogenesis of acne.

BPA, one of the most widely used EDCs, is commonly found in plastic products, food packaging, and thermal paper.<sup>10,11</sup> Due to its lipophilic nature, BPA can be easily absorbed through the skin and accumulate in tissues. Studies have shown that BPA can interact with estrogen receptors, exerting agonistic or antagonistic effects through estrogen receptor (ER)-dependent pathways, and may interfere with IGF-1 activity, thereby affecting sebum secretion and inflammatory responses.<sup>12,13</sup> In addition, BPA can inhibit adiponectin synthesis in adipose tissue, promote the release of pro-inflammatory cytokines, and induce insulin resistance and hyperinsulinemia; the latter may further enhance androgen secretion, forming a vicious cycle that promotes acne development.<sup>14</sup> Both animal and epidemiological studies have indicated that BPA exposure is associated with endocrine disorders such as polycystic ovary syndrome (PCOS) and precocious puberty, which are also closely linked to the occurrence of acne.<sup>15</sup>

Despite these clinical and epidemiological associations, the precise molecular targets and underlying signaling pathways through which BPA exerts its pro-acnegenic effects remain largely unknown. This represents a critical knowledge gap in understanding the environmental etiology of acne. Traditional toxicological approaches are often time-consuming and labor-intensive, making it difficult to systematically elucidate the complex interaction networks among chemicals, proteins, and diseases. In recent years, network toxicology—an integrative discipline that evaluates complex “disease-gene-exposure” interaction networks—has emerged as a novel research strategy that integrates public databases, bioinformatics algorithms, and computational modeling techniques to efficiently predict the potential toxicological mechanisms of environmental chemicals and identify core targets at a systems level.<sup>16,17</sup> This approach has been successfully applied to uncover the molecular mechanisms of BPA in complex diseases such as cardiovascular disorders and autoimmune diseases,<sup>18</sup> demonstrating its strong potential for application.

In this study, BPA was specifically chosen as the focal point over other factors because it is one of the most ubiquitous environmental contaminants with well-documented endocrine-disrupting capabilities, making it an ideal representative model for environmental exposure. To address this knowledge gap, the rationale of our study is to systematically elucidate the molecular mechanisms by which BPA contributes to acne development. This study presents a novel approach by integrating network toxicology with machine learning, molecular docking, and molecular dynamics simulations to decode these complex interactions at a systems level. The findings are expected to provide new scientific evidence for understanding the environmental etiological factors of acne and improving clinical prevention and treatment strategies.

## Methods

### Collection of BPA Targets

The chemical structure and simplified molecular input line entry system (SMILES) formula of BPA were obtained from the PubChem database (<https://pubchem.ncbi.nlm.nih.gov/>). Potential molecular targets of BPA were subsequently identified using the CTD (<https://ctdbase.org/>), SEA (<https://sea.bkslab.org/>), and Swiss Target Prediction (<http://swisstargetprediction.ch/>) databases, restricting the search to Homo sapiens. After integrating the results and removing duplicates, a nonredundant set of BPA-associated targets was established to construct a comprehensive target repository.

## Identification of Potential Acne-Related Targets

The acne-related gene expression dataset GSE6475 was obtained from the Gene Expression Omnibus (GEO) database. This dataset, generated using the Affymetrix Human Genome U133 Plus 2.0 platform, contains transcriptomic profiles from inflammatory acne lesions and matched normal-appearing skin from the same individuals. Prior to differential expression analysis, raw microarray data were preprocessed for background correction and quantile normalization. The probe IDs were mapped to official gene symbols using the corresponding platform annotation files. For multiple probes mapping to the same gene, the average expression value was calculated and retained. All expression data were log2-transformed for subsequent analyses.

## Differential Expression Analysis and WGCNA Module Identification

To identify acne-associated genes, differential expression analysis was performed using the DESeq2 package by comparing transcriptomic profiles between acne lesions and matched normal skin samples. Genes with an adjusted  $p$ -value  $< 0.05$  and  $|\log_2FC| > 1$  were considered significantly dysregulated. Furthermore, Weighted Gene Co-expression Network Analysis (WGCNA) was conducted to identify modules of co-expressed genes. Modules that showed a significant correlation with the acne phenotype ( $P < 0.05$ ) were selected, and all genes within these modules were retained for subsequent analyses.

## Functional Pathway Analysis of Target Genes

Potential target genes mediating the effects of BPA in acne were identified by intersecting BPA-associated targets with acne-related differentially expressed genes (DEGs). The overlapping genes were considered to represent candidates potentially involved in BPA-induced acne pathogenesis. To elucidate their biological significance, Gene Ontology (GO) enrichment analysis—including the categories of biological process (BP), cellular component (CC), and molecular function (MF)—was performed to identify significantly overrepresented terms. In parallel, Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway analysis was conducted to uncover significantly enriched signaling pathways. GO terms and KEGG pathways with an adjusted  $p$ -value  $< 0.05$  were regarded as statistically significant.

## Identification of Core Targets Using Machine Learning Algorithms

To identify key genes associated with BPA-induced acne, two complementary machine learning algorithms—Least Absolute Shrinkage and Selection Operator (LASSO) regression and Support Vector Machine (SVM)—were employed. LASSO is a regularization technique that performs variable selection and prevents overfitting, thereby enhancing model generalizability. SVM, a supervised learning algorithm for classification and regression, was combined with Recursive Feature Elimination (RFE) to iteratively select the optimal subset of features that maximizes model performance. Core targets were defined as the intersection of genes identified by both algorithms. LASSO regression was implemented using the “glmnet” R package (4.4.3) with parameters: `standardize = TRUE`, `alpha = 1`, `family = “binomial”`, and `nfolds = 10`. SVM-RFE analysis was conducted using the “e1071” R package (4.4.3) with linear kernel and automatic scaling of features.

## Gene Set Enrichment Analysis (GSEA) and Mechanistic Exploration

To further elucidate the molecular networks and signaling pathways associated with the core genes, GSEA was performed. This approach evaluates whether predefined gene sets exhibit statistically significant and concordant expression differences between two biological conditions. The core genes identified in the previous analyses were used as the input gene set. GSEA was conducted using reference gene sets from the Molecular Signatures Database (MSigDB), including hallmark and canonical pathway collections. Pathways or biological processes with a normalized enrichment score (NES) and a false discovery rate (FDR)  $q$ -value  $< 0.25$  were considered significantly enriched. This analysis was performed to delineate the key regulatory axes and potential molecular mechanisms through which the core genes might contribute to acne pathogenesis.

## Validation in an Independent Cohort and Immune Infiltration Correlation Analysis

To independently validate the expression profiles of the identified core genes, their transcript levels were analyzed in an external acne dataset, GSE53795, which includes transcriptomic data from acne lesions and matched healthy skin tissues. In addition, to explore the potential immunological relevance of these core genes, immune cell infiltration analysis was performed using the CIBERSORT algorithm with 1000 permutations. This computational method deconvolutes bulk transcriptomic data to estimate the relative proportions of 22 immune cell subtypes. Spearman correlation analysis was subsequently applied to examine the associations between core gene expression levels and the infiltration abundance of specific immune cell populations in acne tissues.

## Molecular Docking Analysis

The three-dimensional crystal structures of the core target proteins, including PLA2G2A, were retrieved from the Protein Data Bank (PDB). The molecular structure of BPA was obtained from the PubChem database in SDF format and converted to PDBQT format using AutoDock Tools (1.5.7). Molecular docking simulations were carried out using AutoDock Vina (1.1.2) to predict the binding affinities and interaction modes between BPA and the target proteins. The docking grid was defined to cover the known active sites of each protein. For each docking run, multiple binding conformations were evaluated, and the conformation with the lowest binding energy was selected as the optimal pose. Key amino acid residues and interaction types were identified in the docking results using PyMOL (2.4.0a0).

## Molecular Dynamics (MD) Simulation

To further assess the structural stability and dynamic behavior of the BPA–PLA2G2A complex, MD simulations were performed using the GROMACS software package (2024.5). The topology parameters for the protein were generated with the CHARMM36 force field, while the ligand parameters for BPA were derived from the CGenFF database. The complex was solvated in a cubic box containing TIP3P water molecules and neutralized by adding counterions to achieve system neutrality. Energy minimization was carried out using the steepest descent algorithm, followed by equilibration under NVT and NPT ensembles. A 100-ns production simulation was then conducted under constant temperature (300 K) and pressure (1 bar) conditions. The stability of the complex was evaluated based on the root mean square deviation (RMSD) and root mean square fluctuation (RMSF) of the protein backbone atoms. Furthermore, hydrogen bond occupancy and other non-covalent interactions throughout the simulation trajectory were analyzed to confirm the persistence and stability of BPA binding as observed in the molecular docking analysis.

## Interaction Mechanism Analysis and Visualization

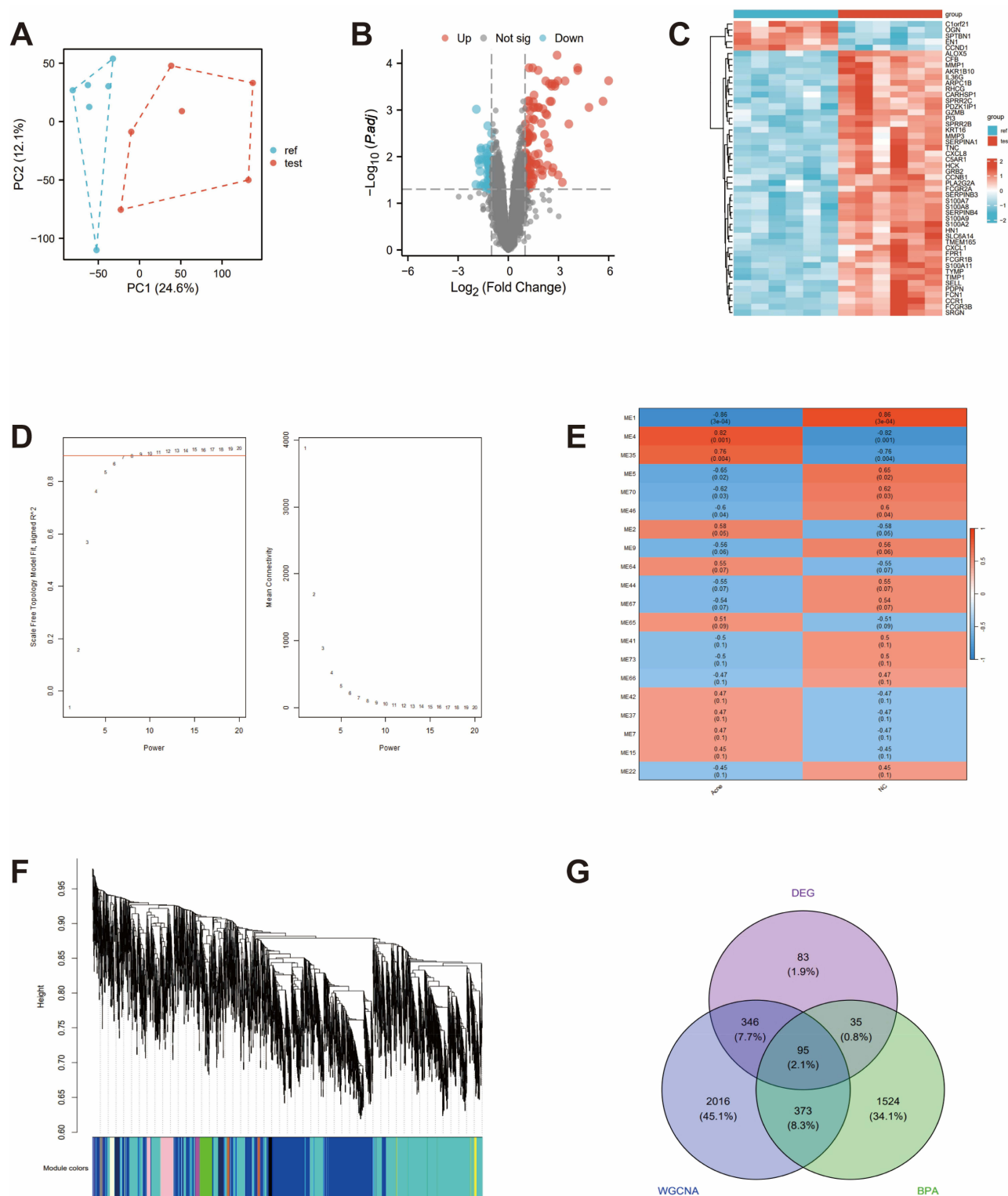
Based on the results of molecular docking and molecular dynamics simulations, the stable binding conformations between BPA and its target proteins (including PLA2G2A) were systematically analyzed. Key intermolecular interactions—such as hydrogen bonding, hydrophobic contacts, and van der Waals forces—were quantitatively characterized and visualized using PyMOL (2.4.0a0). This integrative analysis elucidated the structural determinants underlying the binding stability and specificity of BPA toward its protein targets, providing molecular-level mechanistic insights into their potential interactions.

## Results

### Identification of Potential Targets Involved in BPA-Induced Acne

Target genes associated with BPA were retrieved from integrated public databases and compared with differentially expressed genes identified from human acne tissues (GSE6475) using WGCNA (Figure 1A–F). The overlapping genes were subsequently identified using a Venn diagram (Figure 1G). In total, 95 common targets were obtained and were considered potential genes involved in BPA-induced acne.





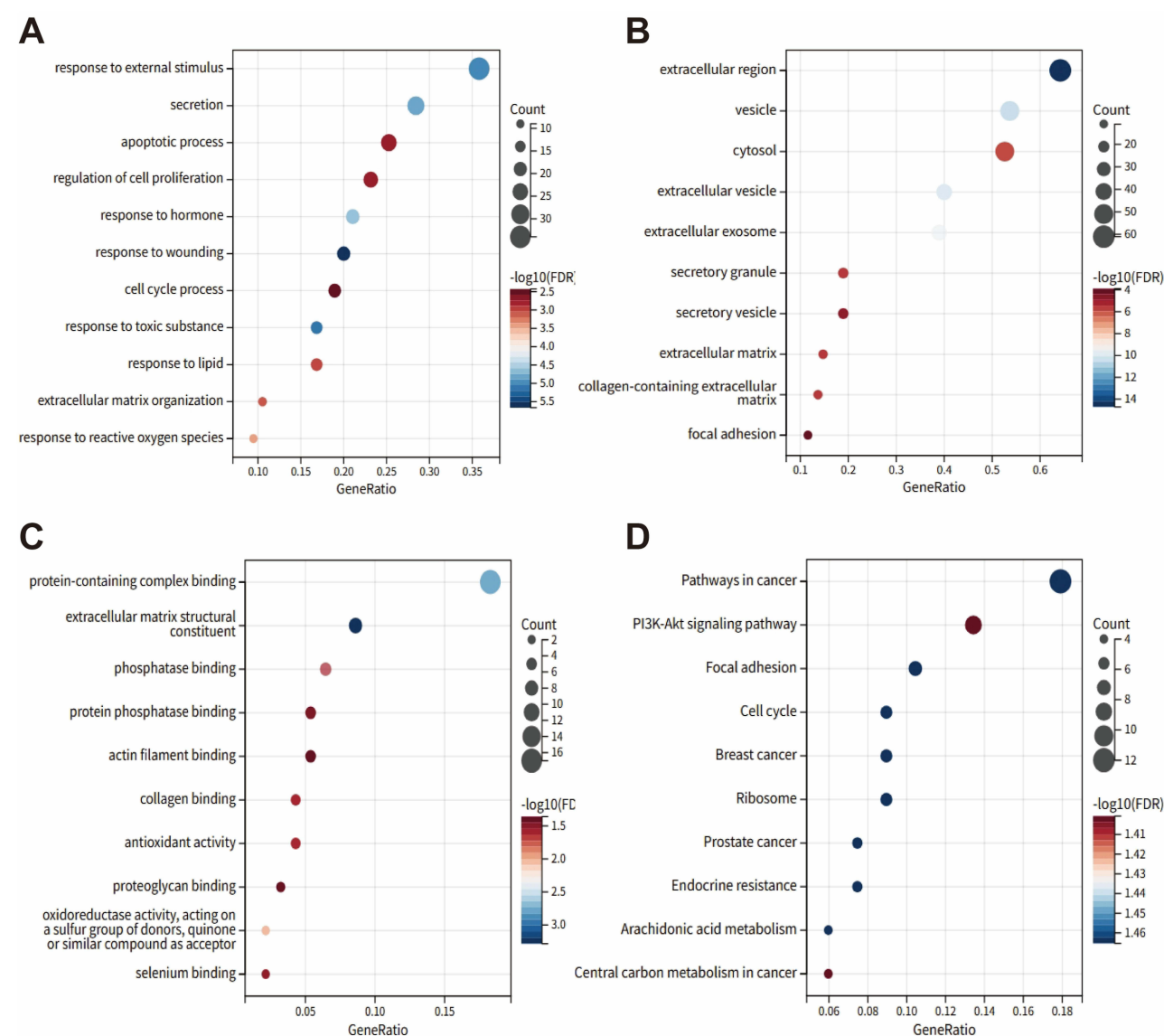
**Figure 1** Identification of potential targets involved in BPA-induced acne. **(A)** Principal component analysis (PCA) of the GSE6475 dataset. **(B)** Volcano plot showing the differentially expressed genes (DEGs). Red dots represent upregulated genes, blue dots represent downregulated genes, and grey dots indicate no significant difference. **(C)** Heatmap of the top DEGs. **(D)** Analysis of network topology for various soft-thresholding powers in WGCNA, displaying the scale-free topology model fit (left) and the mean connectivity (right). **(E)** Heatmap of module-trait relationships. The numbers in each cell represent the correlation coefficient and the corresponding P-value in parentheses. **(F)** Cluster dendrogram of genes based on topological overlap, with assigned module colors. **(G)** Venn diagram illustrating the intersection of DEGs, WGCNA module genes, and BPA-associated targets to identify the 95 common core genes.

## Intersection and Functional Enrichment Analysis

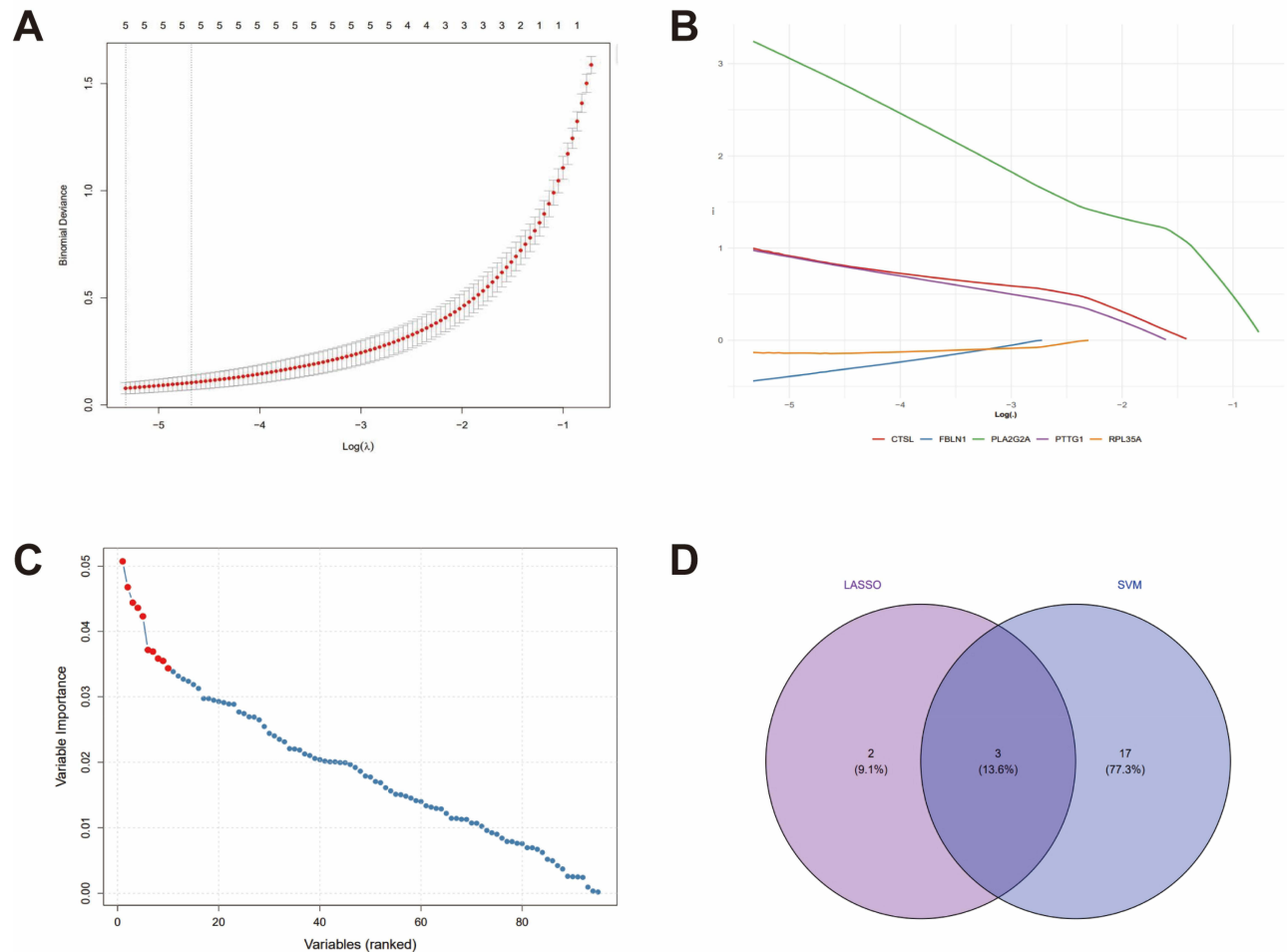
GO (Figure 2A–C) and KEGG (Figure 2D) enrichment analyses were conducted for the 95 intersecting genes. The results showed that these genes were primarily enriched in biological processes related to responses to external stimuli, were mainly localized in the extracellular region, and were significantly associated with the PI3K–AKT signaling pathway, endocrine resistance, and arachidonic acid metabolism pathways (FDR < 0.05).

## Identification of Core Genes Using Machine Learning Algorithms

To further identify core genes involved in BPA-related effects on acne, LASSO regression (Figure 3A and B) and SVM-RFE (Figure 3C) were applied to the intersecting differentially expressed genes for feature selection. Three core genes, CTSL, PTTG1, and PLA2G2A, were ultimately identified (Figure 3D).



**Figure 2** Functional enrichment analysis of the 95 intersecting genes. **(A)** Gene Ontology (GO) Biological Process (BP) enrichment. **(B)** GO Cellular Component (CC) enrichment. **(C)** GO Molecular Function (MF) enrichment. **(D)** KEGG pathway enrichment analysis. The size of the bubbles indicates the number of genes enriched in each term or pathway (Count). The color gradient of the bubbles represents the statistical significance, evaluated by the false discovery rate (FDR) and expressed as  $-\log_{10}(\text{FDR})$ .



**Figure 3** Identification of core genes using machine learning algorithms. **(A)** Ten-fold cross-validation for tuning parameter selection in the LASSO regression model. The red dotted curve represents the binomial deviance, and the optimal  $\lambda$  value was determined based on the minimum criteria. **(B)** LASSO coefficient profiles of the candidate genes plotted against the  $\log(\lambda)$  sequence. **(C)** Feature selection using the Support Vector Machine-Recursive Feature Elimination (SVM-RFE) algorithm. The plot displays the distribution of variable importance for the ranked variables. **(D)** Venn diagram illustrating the intersection of diagnostic markers identified by both the LASSO and SVM algorithms, ultimately yielding 3 core genes (CTSL, PTTG1, and PLA2G2A).

## Enrichment Analysis of Core Genes

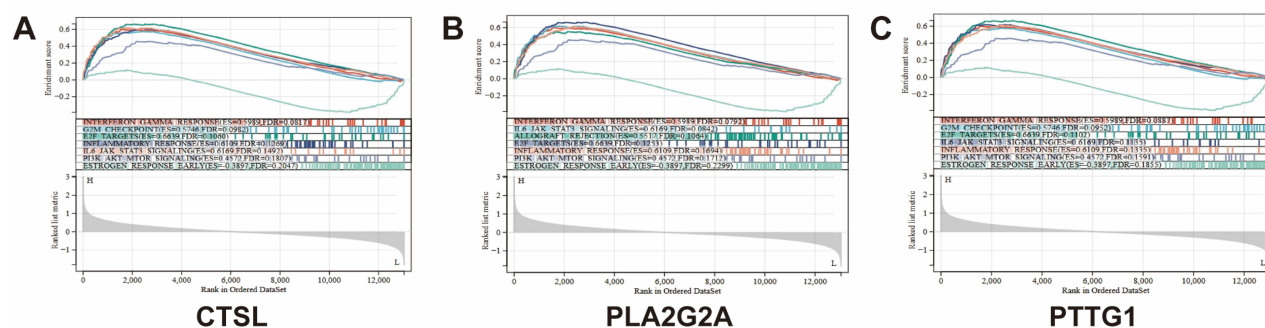
GSEA was performed to further characterize the biological functions and signaling pathways associated with the identified core genes (Figure 4A–C). The results showed that these genes were primarily enriched in pathways related to the interferon- $\gamma$  response, E2F targets, inflammatory response, IL-6/JAK–STAT3 signaling, PI3K–AKT–mTOR signaling, and early estrogen response (FDR < 0.25).

## Validation in an Independent Dataset

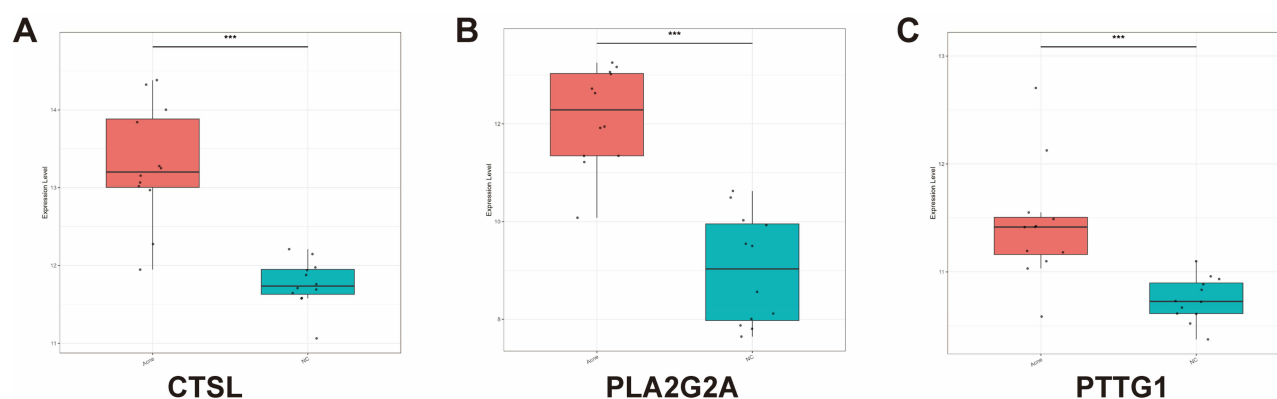
The differential expression of the three core genes, CTSL, PTTG1, and PLA2G2A, was further validated using an independent acne dataset (GSE53795) (Figure 5A–C). The results showed that the expression levels of CTSL, PTTG1, and PLA2G2A were significantly higher in acne tissues compared with normal skin tissues ( $P < 0.001$ ).

## Immune Infiltration Analysis

Immune infiltration analysis revealed a significant increase in pro-inflammatory immune cells, including neutrophils, M1 macrophages, and activated mast cells, in acne tissues ( $P < 0.05$ ). In contrast, regulatory immune cells, such as M2 macrophages and resting dendritic cells, were significantly reduced ( $P < 0.05$ ), indicating an enhanced inflammatory immune state (Figure 6A and C). Correlation analysis further demonstrated that PLA2G2A expression was significantly



**Figure 4** Gene Set Enrichment Analysis (GSEA) of the three identified core genes. (A–C) Multi-GSEA plots illustrating the significantly enriched hallmark pathways and biological processes associated with the expression profiles of (A) CTSL, (B) PLA2G2A, and (C) PTTG1. The top portions of each plot show the enrichment score (ES) curves, the middle portions show the distribution of the core genes within the ranked gene lists, and the bottom portions display the ranked list metrics. The specific ES and false discovery rate (FDR) values for each signaling pathway are provided within the internal legends of the respective plots.



**Figure 5** Validation of the three core genes in an independent dataset. (A–C) Boxplots illustrating the differential expression levels of (A) CTSL, (B) PLA2G2A, and (C) PTTG1 between acne tissues and normal control (NC) skin tissues in the GSE53795 dataset. The statistical significance of the differences between the two groups was evaluated, and the significance levels are denoted by asterisks (\*\*\*) ( $P < 0.001$ ).

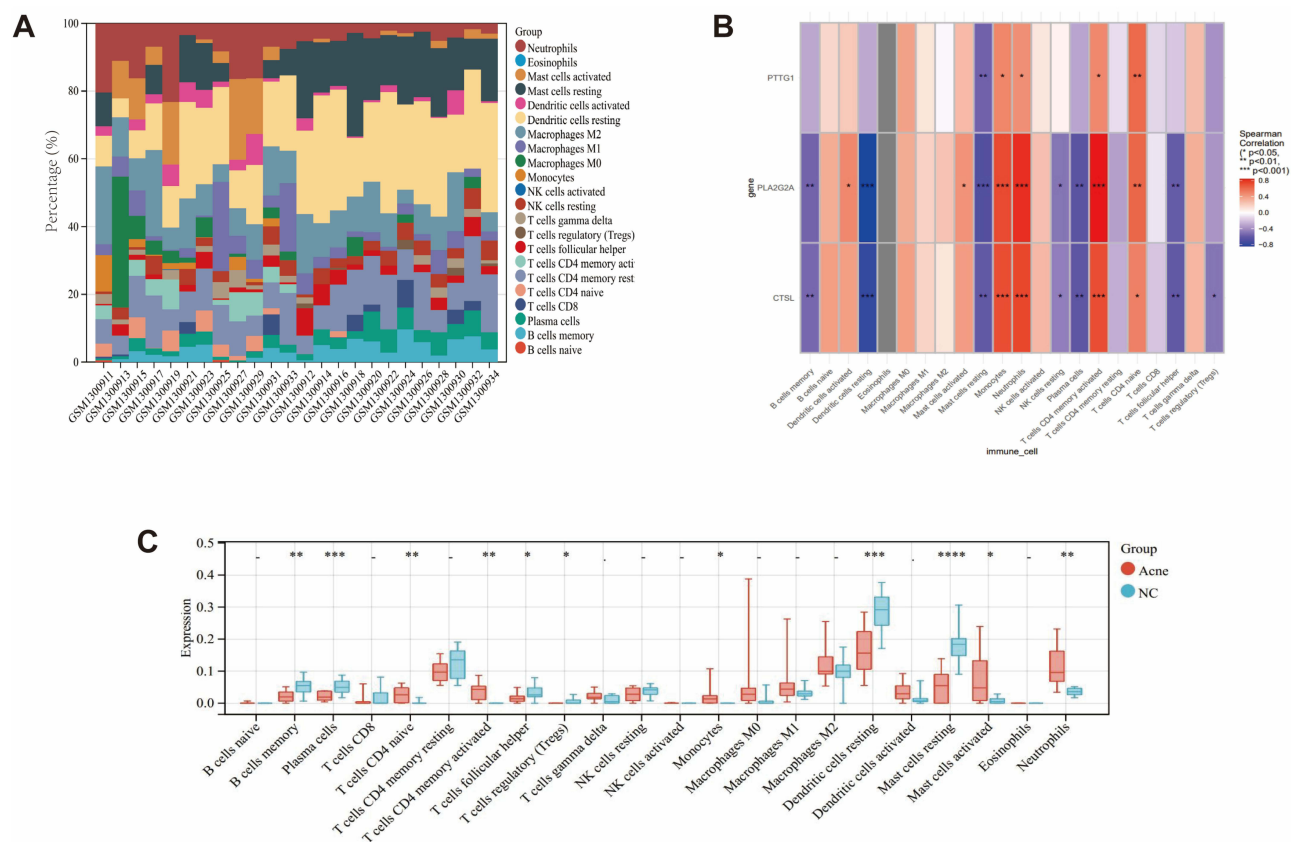
and positively correlated with the infiltration of pro-inflammatory immune cells ( $P < 0.05$ ) and negatively correlated with immune-regulatory cell infiltration ( $P < 0.05$ ) (Figure 6B). These results indicate a potential association between PLA2G2A expression and immune microenvironment alterations in acne.

## Molecular Docking

To evaluate the potential binding interactions between BPA and the identified core targets PLA2G2A, CTSL, and PTTG1, molecular docking analyses were performed. The results showed favorable binding affinities between BPA and all three target proteins, with binding energies consistently below  $-5$  kcal/mol (Table 1), indicating stable molecular interactions. In docking analyses, binding energies below 0 kcal/mol generally reflect spontaneous binding potential, whereas values below  $-5.0$  kcal/mol are commonly considered indicative of relatively strong binding affinity. Visualization of the docking conformations further demonstrated stable binding poses for each BPA–protein complex (Figure 7A–C). Collectively, these results provide structural evidence supporting the potential molecular interactions between BPA and the core acne-related targets identified in this study.

## Molecular Dynamics Simulation

The binding stability of the BPA–PLA2G2A complex was further evaluated using free energy landscape analysis (Figure 8A) and molecular dynamics simulations (Figure 8B). Key dynamic parameters, including the root mean square deviation (RMSD) of the ligand and complex, radius of gyration (Rg), root mean square fluctuation (RMSF), solvent



**Figure 6** Immune infiltration analysis in acne and normal control tissues. **(A)** Stacked bar chart showing the relative abundance of 22 infiltrating immune cell types in each sample. **(B)** Heatmap illustrating the Spearman correlation between the three core genes (PTTG1, PLA2G2A, and CTSL) and the abundance of immune cells. Red cells indicate positive correlations, while blue cells indicate negative correlations. **(C)** Boxplots comparing the differential abundance of the 22 immune cell types between the acne and normal control (NC) groups. The statistical significance levels are denoted by asterisks: \*  $P < 0.05$ , \*\*  $P < 0.01$ , \*\*\*  $P < 0.001$ , and \*\*\*\*  $P < 0.0001$ .

accessible surface area (SASA), and hydrogen bond occupancy, consistently indicated structural stability and conformational compactness throughout the simulation period. These results support the formation of a stable molecular interaction between BPA and PLA2G2A and suggest a potential influence on PLA2G2A-related functional processes relevant to inflammatory signaling in acne.

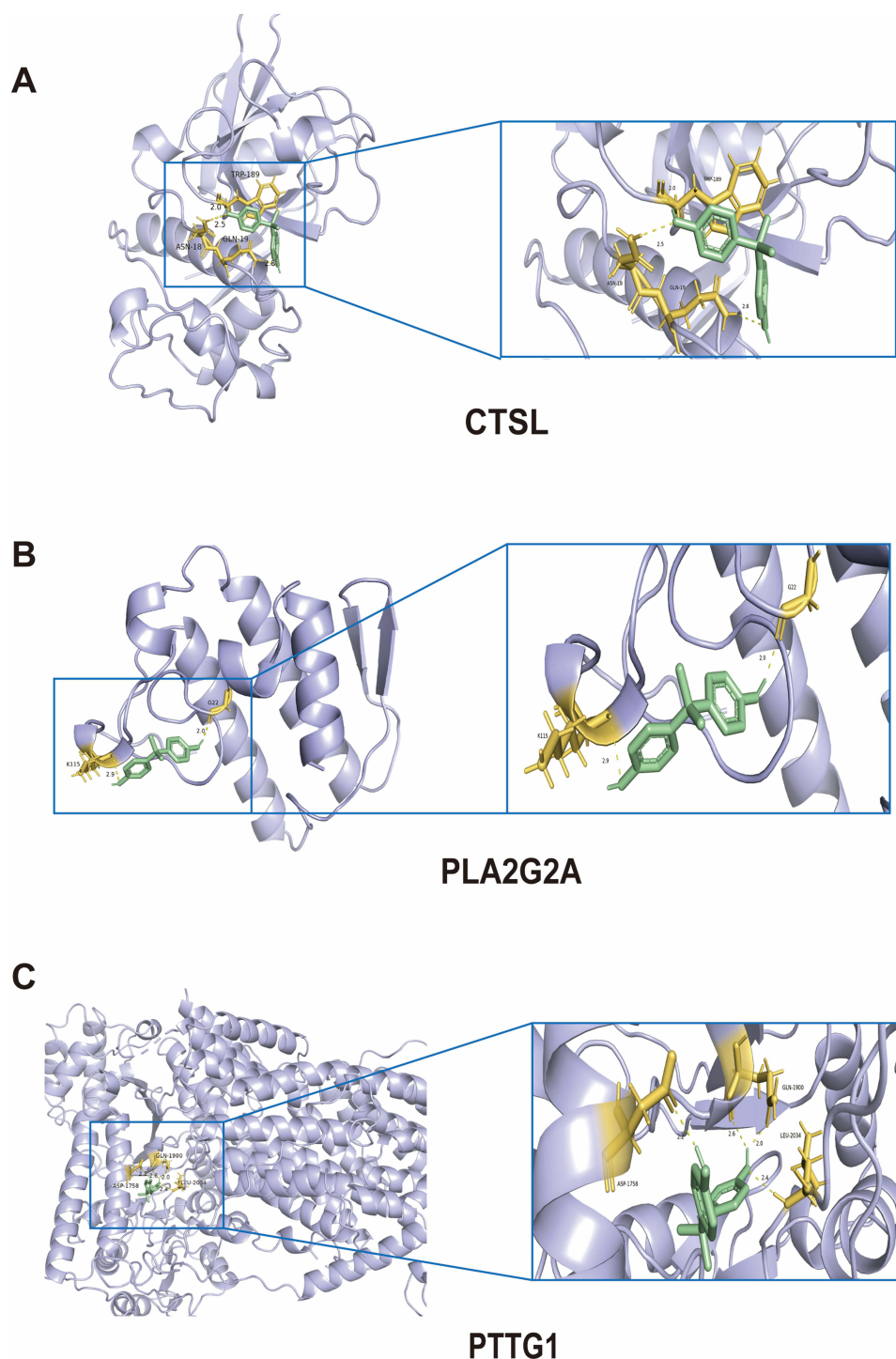
## Discussion

Acne vulgaris is a complex multifactorial skin disorder characterized by the interplay of sebaceous hypersecretion, follicular keratinization, microbial colonization, and dysregulated inflammatory responses. In recent years, increasing attention has been directed toward environmental pollutants as potential aggravating factors in acne development. In the present study, we integrated network toxicology, machine learning, and multidimensional bioinformatics analyses to explore the molecular basis underlying an important epidemiological observation. Previous studies have reported significantly elevated BPA levels in patients with acne, with concentrations positively correlated with disease severity.<sup>19</sup> While traditional literature has primarily attributed BPA's pro-acnegenic effects to its interference with

**Table 1** Binding Energies of Ligands and Receptors

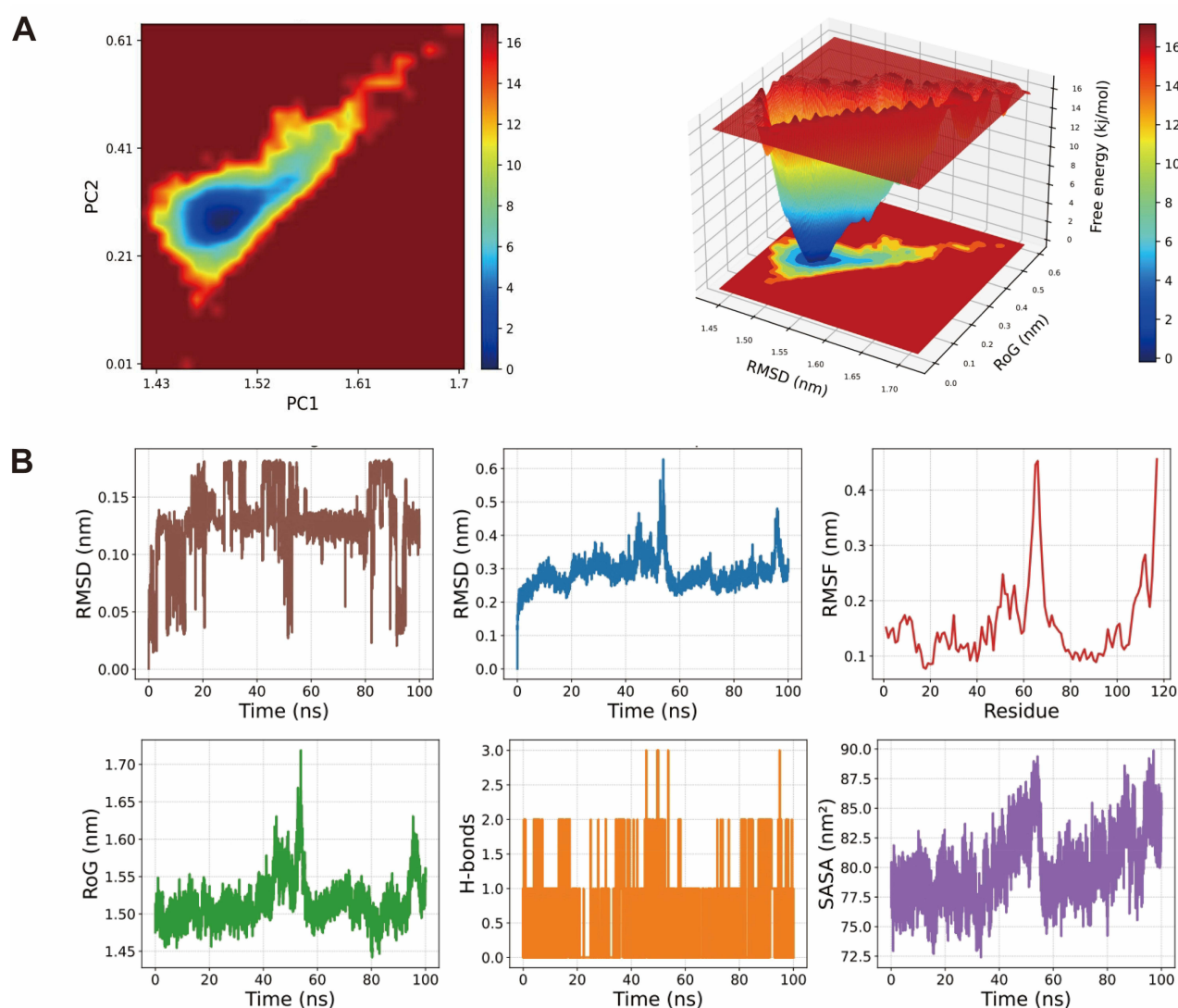
| Ligand | Receptor | Binding Energy [kcal/mol] |
|--------|----------|---------------------------|
| BPA    | CTSL     | -6.07                     |
| BPA    | PLA2G2A  | -5.83                     |
| BPA    | PTTG1    | -6.78                     |





**Figure 7** Molecular docking conformations of BPA with the three core targets. **(A–C)** 3D visualization of the docking complexes for **(A)** CTSL, **(B)** PLA2G2A, and **(C)** PTTG1. In each panel, the left side displays the overall binding pose of BPA within the protein structure, while the right side provides a magnified view of the specific binding pocket. The key interacting amino acid residues are labeled, and the yellow dashed lines represent hydrogen bonds, with the corresponding numbers indicating the hydrogen bond distances in Angstroms (Å).

systemic sex hormones (such as androgen and estrogen receptor signaling),<sup>12,13</sup> our study provides a critical, complementary perspective. We hypothesize that beyond hormonal disruption, BPA exposure may be associated with acne pathogenesis through its direct interaction with three core genes, CTSL, PTTG1, and PLA2G2A, and through the involvement of key signaling pathways, including arachidonic acid metabolism, PI3K–AKT–mTOR, and IL-6/JAK–



**Figure 8** Molecular dynamics simulation of the BPA-PLA2G2A complex. **(A)** Gibbs free energy landscape (FEL) analyses. The 2D contour map (left) is projected onto the first two principal components (PC1 and PC2), and the 3D surface plot (right) displays the free energy (kJ/mol) as a function of the root mean square deviation (RMSD) and radius of gyration (RoG). **(B)** Trajectory analysis of the BPA-PLA2G2A complex over a 100 ns simulation period. The six panels show the RMSD of the ligand, the RMSD of the complex, the root mean square fluctuation (RMSF) of the protein residues, the RoG of the complex, the number of intermolecular hydrogen bonds, and the solvent-accessible surface area (SASA), respectively.

STAT3. These molecular alterations may contribute to immune microenvironment imbalance, thereby facilitating inflammatory processes relevant to acne development and progression.

Preliminary enrichment analyses in this study indicated that the shared targets of BPA exposure and acne were significantly associated with the PI3K–AKT signaling pathway and arachidonic acid metabolism. Previous studies have demonstrated that air pollutants can activate cutaneous inflammation and oxidative stress<sup>20</sup> and are associated with increased acne prevalence.<sup>21–23</sup> Our findings directly link BPA—a specific environmental contaminant—to the core pathological pathways of acne. Notably, we focused on the arachidonic acid metabolism pathway, which plays a crucial role in inflammation and lipid dysregulation, both of which are key mechanisms in acne pathogenesis.<sup>24</sup> Arachidonic acid and its downstream metabolites, such as prostaglandins and leukotrienes, are potent lipid inflammatory mediators that can recruit and activate immune cells, including neutrophils, thereby amplifying local inflammatory cascades.<sup>25,26</sup> Collectively, these observations suggest that BPA exposure may represent an environmental factor influencing arachidonic acid related inflammatory processes relevant to acne.

Through machine learning based feature selection, three core genes, CTSL, PTTG1, and PLA2G2A, were identified as candidates potentially linking BPA exposure to acne related molecular alterations. While all three genes are positively correlated with acne pathogenesis, PLA2G2A was explicitly selected as the central hub for subsequent dynamic simulations and mechanistic exploration for two main reasons. First, biologically, PLA2G2A encodes group IIA phospholipase A2, a key enzyme involved in the hydrolysis of membrane phospholipids and the subsequent release of arachidonic acid, a rate limiting step in eicosanoid biosynthesis and a primary driver of lipid-mediated inflammation in acne.<sup>27</sup> Second, our immune infiltration analysis demonstrated that PLA2G2A exhibited the strongest correlation with key pro-inflammatory immune cells. In the present study, PLA2G2A expression was markedly elevated in acne tissues and showed a positive correlation with the infiltration of pro inflammatory immune cells, including neutrophils and M1 macrophages. These observations are biologically consistent with established inflammatory mechanisms, whereby increased PLA2G2A activity enhances arachidonic acid availability, leading to the production of downstream lipid mediators such as leukotriene B4 that promote neutrophil recruitment and local inflammatory amplification within the pilosebaceous unit.<sup>25,26</sup> Taken together, these findings suggest that BPA exposure may be associated with enhanced PLA2G2A related inflammatory activity, representing a plausible molecular link between environmental exposure and acne associated inflammation.

In addition, the other two core targets, CTSL (cathepsin L) and PTTG1 (pituitary tumor-transforming gene 1), provide further insights into the potential mechanisms of BPA action. CTSL is a lysosomal cysteine protease that participates not only in antigen presentation but also in the direct degradation of extracellular matrix components, thereby contributing to tissue remodeling and the propagation of inflammation. Previous studies have shown that CTSL expression is increased in inflammatory skin diseases such as atopic dermatitis and psoriasis, and that dysregulation of the cystatin M/E–cathepsin axis may impair skin barrier function.<sup>28,29</sup> PTTG1, on the other hand, is an important transcriptional regulator closely involved in cell cycle control and DNA damage repair. The STAT3/PTTG1 signaling axis has been demonstrated to play a crucial role in tumorigenesis,<sup>30</sup> and activation of IL-6/STAT3 signaling can upregulate PTTG1 expression,<sup>31</sup> which is consistent with our GSEA findings showing enrichment of this gene in the IL-6/JAK–STAT3 signaling pathway.

GSEA analysis further reinforced our findings, indicating that the three core genes were collectively associated with the inflammatory response, IL-6/JAK–STAT3, and PI3K–AKT–mTOR signaling pathways. These pathways form a tightly interconnected regulatory network. Specifically, IL-6 may promote T helper 17(Th17) cell differentiation through activation of the JAK–STAT3 pathway,<sup>32</sup> and Th17-related immune inflammatory cascades have been closely linked to acne-associated microbes and disease progression.<sup>33,34</sup> Meanwhile, the PI3K–AKT–mTOR pathway represents a central signaling axis controlling lipid synthesis in sebocytes, and its hyperactivation is a critical cause of seborrhea in acne lesions.<sup>35</sup> Our molecular docking and molecular dynamics simulations further confirmed that BPA could bind to PLA2G2A, CTSL, and PTTG1 with low binding energies and stable conformations. These results provide structural and biophysical evidence supporting the hypothesis that BPA may directly interfere with the functions of these core targets at the molecular level. Moreover, the differential expression of these three genes was further validated in an independent dataset, which also strengthens our mechanistic hypothesis.

In summary, we propose a multi step mechanistic model to describe how BPA exposure may contribute to acne development. As a putative molecular initiating event, BPA may interact with the core targets PLA2G2A, CTSL, and PTTG1, thereby initiating a series of downstream molecular and cellular responses. On one hand, BPA associated upregulation of PLA2G2A may enhance arachidonic acid metabolism, leading to increased production of pro inflammatory lipid mediators and the recruitment of neutrophils and other inflammatory immune cells, which collectively amplify local inflammatory responses. On the other hand, through potential modulation of PTTG1 and CTSL, BPA exposure may influence the PI3K–AKT–mTOR and JAK–STAT3 signaling pathways, resulting in altered sebaceous activity and immune regulation. Together, these interconnected processes may act synergistically to promote the initiation and progression of acne.

Despite its strengths, this study has several limitations. First, it must be acknowledged that the current findings are primarily exploratory and derived from bioinformatic analyses and computational simulations. Although validation was performed using an independent dataset, further *in vitro* and *in vivo* experimental studies are imperative to confirm the biological relevance of these findings. Specifically, future research must investigate how the BPA-PLA2G2A interaction

affects the arachidonic acid metabolic pathway and subsequent inflammation at the cellular level, and comprehensively analyze the post-interaction metabolic changes (such as shifts in downstream lipid mediators) to fully validate this mechanistic axis. Second, the absence of longitudinal or prospective cohort data limits the ability to directly associate *in vivo* BPA exposure levels with dynamic changes in core gene expression and clinical disease progression. Moreover, acne pathogenesis is shaped by a complex interplay of environmental exposures, genetic susceptibility, and lifestyle factors, which together influence disease onset and severity. Future studies integrating multi-omics profiling with exposome assessment and detailed clinical phenotyping will be essential to further elucidate the systemic and interactive mechanisms underlying BPA-associated acne.

Nevertheless, to our knowledge, this study is among the first to systematically explore the association between BPA exposure and acne pathogenesis by integrating multi omics analyses with molecular docking and dynamics simulations, with particular emphasis on the PLA2G2A arachidonic acid inflammation axis. This work provides a novel mechanistic perspective on how environmental endocrine disruptors may contribute to inflammatory skin diseases and highlights the value of combining computational toxicology with dermatological research. From a public health and clinical standpoint, these findings underscore the importance of considering environmental exposure factors, such as BPA, in the etiological assessment of acne. They further suggest that strategies aimed at reducing exposure to endocrine disrupting chemicals may represent a complementary approach for the prevention and comprehensive management of acne and related inflammatory skin conditions. Furthermore, given the central role of PLA2G2A in the lipid-driven inflammatory cascade and its stable interaction with BPA, targeting this specific axis could offer novel, precision-based therapeutic strategies for patients with refractory acne linked to environmental exposures. Specific approaches may include the development of topical PLA2G2A inhibitors or the use of non-steroidal agents to modulate downstream arachidonic acid metabolism.

## Conclusion

This study identified CTSL, PTTG1, and PLA2G2A as core target genes potentially linking BPA exposure to acne pathogenesis. Among these genes, PLA2G2A emerged as a central candidate associated with arachidonic acid metabolism. Molecular docking and molecular dynamics simulations supported the formation of a stable interaction between BPA and PLA2G2A, suggesting a potential molecular basis for BPA related inflammatory effects. Overall, while this study is exploratory and primarily based on computational modeling, it provides valuable evidence supporting the role of BPA as a potential environmental risk factor for acne. From a public health and clinical perspective, these findings underscore the importance of mitigating environmental endocrine disruptors in dermatological care. Crucially, the structural stability demonstrated in our molecular dynamics simulations strongly supports PLA2G2A as a viable, druggable target. This highlights the potential of developing topical PLA2G2A inhibitors as a novel, precision-based treatment strategy for environmentally exacerbated acne. Finally, this axis warrants further rigorous *in vitro* and *in vivo* experimental validation to fully elucidate its clinical and therapeutic potential.

## Data Sharing Statement

The datasets generated and/or analyzed during the current study are available from the corresponding author upon reasonable request.

## Ethics Declaration

This study is a secondary analysis based on legally obtained, publicly available, and de-identified data from public databases. It does not involve any direct interaction with human subjects or the use of identifiable private information. According to Items 1 and 2 of Article 32 of the Measures for Ethical Review of Life Science and Medical Research Involving Human Subjects (issued by the National Health Commission of the People's Republic of China on February 18, 2023), this research meets the criteria for exemption from institutional ethical review.

## Author Contributions

All authors made a significant contribution to the work reported, whether that is in the conception, study design, execution, acquisition of data, analysis and interpretation, or in all these areas; took part in drafting, revising or critically



reviewing the article; gave final approval of the version to be published; have agreed on the journal to which the article has been submitted; and agree to be accountable for all aspects of the work.

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## Disclosure

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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